

MASTER BUILD DOCUMENT • V1.0

DRISHTI



The world's first Hindi-native Industrial AI Agent. A complete product specification, technology blueprint, and build guide for an AI safety platform that solves 40 confirmed problems at IFFCO Phulpur.

PRODUCT

DRISHTI Platform

CATEGORY

Industrial AI SaaS

TARGET

IFFCO Phulpur

DATE

June 2026

CONFIDENTIAL — INTERNAL USE

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CHAPTER 01

Executive Summary & Vision

What DRISHTI is, who it serves, and why it matters

One-line pitch: DRISHTI is the world's first Hindi-native Industrial AI Agent that protects field engineers, predicts plant failures before they happen, and lets engineers talk to their plant in natural language — solving 40 confirmed problems that killed four IFFCO engineers and cost crores in production losses.

The core insight

Yokogawa has spent fifty years making the *process* smarter through CENTUM VP, Exaquantum, and PRM. Nobody made the *person* smarter or safer. DRISHTI fills that gap. Where the DCS shows "what is happening," DRISHTI explains "why it is happening, what will happen next, and what to do" — in Hindi.

ATTRIBUTE	VALUE
Product name	DRISHTI — Vision (Hindi)
Category	Industrial AI SaaS · Safety Intelligence Platform
Target users	Field engineers, shift operators, I&C technicians, plant managers
Market path	IFFCO Phulpur → 5 IFFCO plants → 34 Indian urea plants → Global
What Yokogawa does	Monitors the process only
What DRISHTI does	Protects the person, predicts failures, speaks Hindi, learns across plants
Revenue model	SaaS — Rs 5-15 lakh per plant per year
IP ownership	100% founder-owned (register OPC before demo)
Build timeline	4-week MVP · 3-month full product
Core AI engine	Claude (claude-sonnet-4) + Whisper + YOLOv8 + PyTorch

Why this wins awards: An instrumentation intern at IFFCO Phulpur — the same plant where four engineers died in 2020-21 — identified the exact gap that a USD 3.3 billion company with fifty years of experience missed, and built the solution in one month. In Hindi. On a phone. That story wins every engineering innovation award in India.

CHAPTER 02

The 40 Confirmed Problems

Every problem is sourced — use these in engineer meetings and award applications

LIFE SAFETY — 8 Problems

P-01 No field engineer location tracking

Dec 2020, IFFCO Phulpur — 2 engineers died. No real-time location system existed. (Source: UP CM probe order)

P-02 No dead man's switch for lone workers

Night-shift engineers enter hazardous zones alone with zero automatic check-in. (Source: incident pattern 2018-2021)

P-03 No pre-entry zone hazard intelligence

Engineers enter zones without knowing active alarms, bypasses, or gas levels. (Source: both 2020 & 2021 incidents)

P-04 No NH3 gas alert to engineer's mobile

Gas sensors alert only the HIS screen. Field engineer near a leak gets no personal warning. (Source: leak pattern)

P-05 Paper SIS bypass — IEC 61511 violation

IEC 61511 requires digital bypass management. Paper cannot meet this. IFFCO is non-compliant at all plants.

P-06 No simultaneous bypass conflict detection

Multiple bypasses on related interlocks = catastrophic risk. Paper cannot detect this. (Source: IEC 61511, ISA 84)

P-07 Slow emergency response — manual calls only

No one-tap SOS, no auto-GPS, no simultaneous supervisor alert. Dec 2020 response was delayed.

P-08 No incident pattern prediction

5 incidents in 4 years at Phulpur. No AI matches current conditions to pre-incident signatures. (Source: IFFCO records)

EQUIPMENT & MAINTENANCE — 9 Problems

P-09 Rotating equipment fails without warning

Dec 2020: ammonia pump tie rods broke → 2 deaths. Detectable weeks earlier via vibration. (Source: incident report)

P-10 No vibration trend analysis

Bently Nevada data exists but no analytics dashboard. Bearing failures are surprises.

P-11 Boiler health monitoring inadequate

Mar 2021: boiler explosion → 2 deaths. 3 months after pump incident, same plant. (Source: news reports)

P-12 Calibration records on paper registers

No drift trending, no due-date alerts, no digital certificates. (Source: standard Indian PSU practice)

P-13 No instrument drift trend analysis

As-found/as-left data on paper is never analyzed. Transmitter failures are surprises.

P-14 HART device diagnostic data not analyzed

CENTUM VP PRM collects HART health data; zero analytics built on it. (Source: Yokogawa PRM docs)

P-15 Phulpur-I legacy 4-20mA — 45 years old

No remote diagnostics. (Source: IFFCO MoEF filing — Rs 976 Cr upgrade project filed)

P-16 Control valve stiction not tracked

HP urea section valves develop stiction; detected only when operators notice loop oscillation.

P-17 No Remaining Useful Life calculation

No equipment life estimation. Industry: RUL reduces downtime 25-35%. (Source: predictive maint. research)

DAILY OPERATIONS — 10 Problems

P-18 Daily reports printed on paper

CONFIRMED: HP printer labeled "For Daily Reports" photographed in Phulpur control room, June 2026.

P-19 Paper shift logbook (green notebook)

CONFIRMED: green notebook on operator desk in Phulpur control room photo, June 2026.

P-20 Verbal shift handover — information lost

(Source: ISA blog — Reliance SVP documented operating windows in Excel/Word/PDF)

P-21 Alarm flood — ISA-18.2 non-compliant

Yokogawa: "poor alarm management caused USD 80M damage, 26 injured." (Source: Yokogawa docs)

P-22 Top bad-actor alarms never identified

10 tags cause 60%+ of all alarms; never fixed at IFFCO. (Source: ISA-18.2 research)

P-23 PID loops left in manual mode

After upsets, operators forget to restore AUTO. Plant runs inefficiently for hours.

P-24 No mobile dashboard for management

Exaquantum accessible only from plant intranet. (Source: Exaquantum technical docs)

P-25 No Hindi language interface in DCS

Yokogawa has never built a Hindi product in 50 years. (Source: complete product review)

P-26 Knowledge lost when engineers retire

45-year-old plants; original engineers retiring; no knowledge capture system.

P-27 No cross-plant knowledge sharing

Phulpur solutions never reach Aonla/Kalol/Kandla. Same problems repeat across plants.

DATA & CYBERSECURITY — 6 Problems

P-28 JIFCO Jordan ransomware — confirmed breach

CONFIRMED: Apr 2026 ransomware "netrunner" hit IFFCO JV; systems encrypted, data stolen.

P-29 Exaquantum locked in control room only

(Source: Saint-Gobain India documented this exact Yokogawa OPC-DA vs OPC-UA conflict)

P-30 No supply chain visibility — Paradeep shutdown

CONFIRMED: May 2026 complete Paradeep shutdown; Strait of Hormuz cut supply. (Source: Parliament records)

P-31 No OT cybersecurity monitoring

No Claroty/Dravos equivalent at any IFFCO plant. JIFCO breach confirms active threat.

P-32 DCS data not integrated with SAP

Manual production accounting; discrepancies common. (Source: ERP-DCS integration research)

P-33 Two DCS systems — data not unified

Phulpur-I: CENTUM CS 3000; Phulpur-II: CENTUM VP. No unified view.

ENERGY & YOKOGAWA-SPECIFIC GAPS — 7 Problems

P-34 Energy monitored in silos — not integrated

(Source: Yokogawa Apr 2026 + IFFCO FY22-23 report: 5.251 vs 5.200 Gcal/MT target)

P-35 Reformer combustion not continuously optimized

Energy is 70-80% of urea production cost. (Source: IFFCO MoEF energy filings)

P-36	Steam balance not monitored holistically Per-section monitoring misses venting losses. (Source: Yokogawa energy research 2026)
P-37	CENTUM VP shows WHAT — not WHY or NEXT No predictive intelligence, no root cause. (Source: CENTUM VP product docs)
P-38	No conversational interface Yokogawa never built NLP in 50 years. (Source: complete portfolio analysis)
P-39	No mobile app for field operations CENTUM VP needs a HIS workstation; no mobile product exists.
P-40	No operator training simulator New operators learn on live plant; no OTS deployed at IFFCO.

CHAPTER 03

Complete Solutions Map

Every problem mapped to the DRISHTI module that fixes it

PROBLEMS	SEVERITY	DRISHTI MODULE	HOW IT IS FIXED
P-01 to P-03 (tracking/hazard)	Critical	Guardian Core	GPS zone tracking + QR hazard brief before entry
P-04 (NH3 alert)	Critical	Gas Alert	Sensor API to real-time mobile push
P-05 to P-06 (bypass)	Critical	SafePermit	Digital workflow + conflict detection engine
P-07 (emergency)	Critical	SOS Module	One-tap SOS to GPS + auto-call all supervisors
P-08 (prediction)	High	IncidentAI	Historical pattern matching vs current conditions
P-09 to P-11 (equipment)	Critical	EquipmentIQ	CUSUM + LSTM vibration trends, 2-8 week warning
P-12 to P-13 (calibration)	High	CalTrack	Digital records + drift analytics + PDF certs
P-14 (HART)	High	HART Analytics	Parse PRM data to device health dashboard
P-18 to P-20 (paper)	High	ShiftLog	Digital logbook + auto report + WhatsApp delivery
P-21 to P-22 (alarms)	High	AlarmIQ	ISA-18.2 KPI engine + bad-actor ranking
P-25 (Hindi)	High	PlantGPT Hindi	Whisper to Claude to ElevenLabs Hindi TTS
P-26 to P-27 (knowledge)	Medium	KnowledgeRAG	Voice-log to LlamaIndex + Pinecone searchable KB
P-28 to P-31 (cyber)	Critical	OT Security	Network anomaly detection + audit logging
P-34 to P-36 (energy)	High	EnergyIQ	Unified dashboard + Claude optimization suggestions
P-37 to P-39 (Yokogawa gaps)	High	PlantGPT + 3D Twin	Conversational AI + Three.js plant visualization

CHAPTER 04

Full Technology Stack (2026)

Every framework, library, and tool — complete startup-grade selection

Web Frontend

- Next.js 15 (App Router)
- React 19
- TypeScript 5.x
- Tailwind CSS 4
- shadcn/ui + Radix
- Framer Motion 11
- Three.js r165
- React Three Fiber
- D3.js + Recharts
- Zustand
- TanStack Query v5
- socket.io-client

Mobile App

- React Native 0.75
- Expo SDK 53
- Expo Router
- Reanimated 3
- Expo Camera (QR)
- Expo Audio
- Web Speech API (Hindi)
- Expo Notifications
- React Native Maps
- MMKV
- WatermelonDB (offline)

Backend API

- FastAPI (async)
- Python 3.12
- Pydantic v2
- SQLAlchemy 2.0 async
- Alembic
- python-socketio
- Celery 5.x
- APScheduler
- JWT + passlib
- Twilio
- ReportLab
- WhatsApp Business API

AI / ML Brain

- Claude claude-sonnet-4
- Anthropic SDK
- LangChain 0.3
- LlamaIndex 0.11
- Pinecone (vectors)
- Whisper v3 (Hindi STT)
- ElevenLabs (Hindi TTS)
- scikit-learn
- PyTorch + LSTM
- YOLOv8
- librosa (sound FFT)
- DWSIM (simulation)

Data Infrastructure

- TimescaleDB
- PostgreSQL 16
- Redis 7
- Apache Kafka
- MinIO / S3
- Pinecone
- SQLite (offline)
- opcu (OPC-UA)
- OpenOPC (OPC-DA)
- pandas
- numpy + scipy
- ONNX

DevOps & Infrastructure

- Turborepo
- pnpm workspaces
- Docker + Compose
- GitHub Actions
- Vercel (web)
- Railway (API)
- Nginx
- Sentry
- Prometheus + Grafana
- Vitest + Playwright
- ESLint + Biome

System Architecture

How data flows from the DCS to the engineer's phone

LAYER	PURPOSE	KEY TECHNOLOGIES
Data Sources	What DRISHTI reads	Yokogawa CENTUM VP (OPC-UA), Exaquantum (OPC-HDA), CCTV, phone microphone, weather API, CSV exports
Layer 1 — Ingestion	Collect & store data	opcua client polling every 5s, pandas CSV parser, Kafka producer, TimescaleDB write, Pydantic validation
Layer 2 — Analytics	Detect & predict	CUSUM drift detection, Isolation Forest anomaly, LSTM 2-hr prediction, FFT sound analysis, ISA-18.2 KPI engine, YOLOv8
Layer 3 — AI Brain	Reason & explain	Claude (claude-sonnet-4) reasoning, LangChain orchestration, LlamaIndex + Pinecone RAG, DWSIM simulation, Whisper Hindi STT
Layer 4 — Interfaces	How engineers interact	React Native mobile, Next.js web dashboard, WhatsApp bot, SMS (Twilio), 3D Digital Twin, Hindi voice (ElevenLabs)

Key principle: DRISHTI reads *from* Yokogawa but never depends on it. It works even if the DCS interface is down. There is no Yokogawa equivalent, so they cannot offer "use our product instead."

CHAPTER 06

Complete Feature Specification

Every DRISHTI feature in detail

Dead Man's Switch + Zone Tracking

Phase 1

Engineer enters hazardous zone, mobile activates a 15-minute check-in timer. No check-in fires SMS + auto-call to supervisor + control room alert. QR code at zone entrance shows active alarms, SIS bypasses, NH3 levels, and last incident. GPS tracks engineer location throughout. Solves P-01, P-02, P-03.

One-Tap Emergency SOS

Phase 1

Single large button sends GPS coordinates + zone name + last 10 process readings, then calls supervisor + safety officer + control room simultaneously and logs the incident timeline automatically. Response time falls from 5 minutes to under 30 seconds. Solves P-07.

PlantGPT — Hindi Voice AI

Phase 1

Whisper v3 converts Hindi speech to text, LangChain formats plant context, Claude reasons and answers, ElevenLabs speaks the answer in Hindi. The engineer asks hands-free in the field and gets a specific, data-driven answer. First Hindi industrial AI ever built. Solves P-25, P-37, P-38.

SafePermit — Digital Bypass Management

Phase 1

IEC 61511 compliant digital permit-to-work. Engineer requests a bypass, supervisor approves with a digital signature, the system sets an expiry timer, and a live dashboard shows every active bypass. A risk-matrix engine detects dangerous simultaneous combinations and auto-escalates expired bypasses. Full legal audit trail. Solves P-05, P-06.

EquipmentIQ — Predictive Maintenance

Phase 2

Technicians log vibration, temperature, and oil pressure during rounds. CUSUM detects sustained drift, an LSTM predicts failure 2-8 weeks out, and Isolation Forest catches unexpected anomalies. Point a phone microphone at a pump for librosa FFT sound diagnosis: "Cavitation detected, failure in ~12 days." Auto-generates a work order. Solves P-09 to P-17.

Digital ShiftLog + Auto Report

Phase 2

Replaces the green notebook (confirmed in your photo). Operators fill a structured digital shift form; AI auto-generates the complete handover summary and a PDF report from DCS data plus manual entries, delivered to the supervisor by WhatsApp. Replaces the "For Daily Reports" printer (confirmed in your photo). Solves P-18, P-19, P-20.

AlarmIQ — ISA-18.2 Alarm Analytics

Phase 2

Upload an Exaquantum alarm CSV; Python identifies the top 10 bad-actor tags, calculates flood frequency, maps stale alarms, and generates an ISA-18.2 KPI compliance PDF with a clear action list: "Fix these 10 tags to reduce 60% of alarm volume." Solves P-21, P-22.

CalTrack — Calibration Management

Phase 2

Full instrument tag database. Technician enters as-found/as-left values on mobile; the system computes drift, generates a PDF certificate, and sends due-date alerts (30/7/1 days). Drift-trend charts predict the next calibration date. Replaces paper calibration registers. Solves P-12, P-13.

IncidentAI — Pattern Predictor

Phase 3

Loads IFFCO's historical incidents (Dec 2020 pump, Mar 2021 boiler, ammonia leaks). Extracts pre-incident signatures and compares current conditions using cosine similarity. Alerts above 60%: "Current P-201 conditions match Dec 2020 incident by 73% — inspect now." Claude explains the pattern in Hindi. Solves P-08.

3D Plant Digital Twin

Phase 3

A Three.js 3D model of plant sections with live DCS values overlaid via WebSocket. Equipment is colour-coded (green/amber/red). Click any asset to open a health panel with trend charts and a DRISHTI AI comment. The visual showstopper for the demo and award. Solves P-37.

EnergyIQ — AI Energy Optimizer

Phase 3

Unified view of gas + steam + power as one Gcal/MT metric versus the 5.200 target. Claude suggests specific setpoint changes with ROI: "Increase FRC-101 by 2% to save 0.12 Gcal/MT — about Rs 18 lakh/month at current gas price." Solves P-34, P-35, P-36.

KnowledgeRAG — Plant Knowledge Base

Phase 3

Engineers voice-log field observations in 30 seconds. LlamaIndex + Pinecone make it semantically searchable. Ask later, "P-101 pump ke liye pichli baar kya kiya tha?" and DRISHTI retrieves the exact solution. Thirty years of knowledge, never lost; shared across plants. Solves P-26, P-27.

CHAPTER 07

Phase-Wise Build Plan

What gets built each week — realistic and achievable

WEEK 1 — DAYS 1-7 Foundation + Voice AI	
✓ Turborepo monorepo setup ✓ Next.js 15 + FastAPI scaffold ✓ TimescaleDB + Redis setup ✓ Claude API integration ✓ Whisper Hindi STT	✓ ElevenLabs Hindi TTS ✓ PlantGPT basic conversation ✓ JWT auth + user roles ✓ React Native Expo setup ✓ Mock DCS data seeding
WEEK 2 — DAYS 8-14 Safety + Maintenance	
✓ Dead Man's Switch (mobile) ✓ Zone QR hazard brief ✓ One-tap SOS + Twilio ✓ SafePermit bypass workflow ✓ CUSUM anomaly engine	✓ CalTrack mobile entry ✓ Digital ShiftLog ✓ Auto shift report PDF ✓ WhatsApp alert delivery ✓ Bypass risk matrix
WEEK 3 — DAYS 15-21 Intelligence + Prediction	
✓ LSTM time-series model ✓ AlarmIQ CSV analytics ✓ ISA-18.2 KPI report ✓ Sound diagnostic (librosa) ✓ IncidentAI pattern match	✓ EnergyIQ dashboard ✓ LlamaIndex + Pinecone RAG ✓ 3D Digital Twin (Three.js) ✓ YOLO CCTV analysis ✓ Management mobile view
WEEK 4 — DAYS 22-30 Polish + Demo	
✓ Real IFFCO data integration ✓ Docker Compose full stack ✓ GitHub Actions CI/CD ✓ Vercel + Railway deploy ✓ IFFCO theme + branding	✓ Demo scenario scripting ✓ Engineer demo walkthrough ✓ Award application write-up ✓ Pitch deck (Figma) ✓ OPC company registration

CHAPTER 08

Build Prompts — Week 1: Foundation

Paste each prompt into Claude Code in order

How to use: Open your terminal, run `claude` in your project folder, and paste these prompts one at a time. Complete and verify each before moving to the next.

PROMPT 1 — PROJECT INITIALIZATION

Initialize a Turborepo monorepo called "drishti" for an industrial AI safety platform for IFFCO fertilizer plant.

Structure:

- apps/web — Next.js 15 with TypeScript, Tailwind CSS 4, App Router, shadcn/ui
- apps/mobile — React Native with Expo SDK 53, TypeScript, Expo Router
- apps/api — FastAPI Python 3.12 with async support
- packages/types — Shared TypeScript types
- packages/ui — Shared UI components

Use pnpm workspaces. Install all dependencies. Create a dark industrial theme with green (#00C864) as primary accent. Product name is DRISHTI. Target users are field engineers at IFFCO Phulpur ammonia/urea plant. Run dev servers and confirm everything works.

PROMPT 2 — DATABASE SETUP

In apps/api, set up the complete database layer:

1. Install asyncpg, sqlalchemy[asyncio], alembic, timescaledb support
2. PostgreSQL + TimescaleDB connection for time-series DCS data
3. SQLAlchemy models: User (role: ENGINEER/SUPERVISOR/MANAGER), Zone (hazard_level), BypassPermit, ShiftLog, Instrument, CalibrationRecord, ProcessData (TimescaleDB hypertable), Incident, EquipmentHealth
4. Alembic migrations
5. Seed realistic IFFCO Phulpur mock data: 30 instruments, 3 shifts, 48 hours of process data
6. Redis connection for caching and pub/sub

PROMPT 3 — HINDI PLANTGPT AI CORE

Build the PlantGPT Hindi AI system in apps/api:

1. Install anthropic, openai (Whisper), python-multipart
2. /api/ai/chat endpoint: accept text OR audio; if audio, transcribe with Whisper language="hi"; fetch last 2 hours process data + active alarms + bypasses as context; call Claude claude-sonnet-4 with system prompt establishing DRISHTI persona that answers in the user's language with specific numbers; return text + suggested_actions
3. /api/ai/voice-response: text to Hindi audio via ElevenLabs (gTTS fallback)
4. Test queries: "Aaj raat kya problem aa sakti hai?", "P-101 pump ki temperature kya hai?"

PROMPT 4 — WEB DASHBOARD HOMEPAGE

Build the DRISHTI web dashboard homepage in apps/web (dark theme, green #00C864, Syne headings, DM Sans body):

1. Overview page: top bar with live clock + user; 4 KPI cards (Urea MT, Energy Gcal/MT, Active Alarms, Active Bypasses) with green/amber/red status; 24-hour production trend (Recharts); unacknowledged alarms table; field engineers-in-zones count; real-time via WebSocket
 2. Sidebar nav: Overview, Field Safety, SafePermit, AlarmIQ, CalTrack, ShiftLog, PlantGPT, Settings
 3. PlantGPT floating chat button with Hindi voice input
- Connect WebSocket to ws://localhost:8000/ws/dashboard

PROMPT 5 — MOBILE APP + DEAD MAN'S SWITCH

Build DRISHTI mobile app core in apps/mobile (Expo):

1. Expo Router: (auth)/login, (app)/index, (app)/zone-entry (QR), (app)/dead-mans-switch, (app)/sos, (app)/plantgpt
2. Dead Man's Switch: shows current zone + 15-min countdown (MM:SS); large green CHECK IN button resets timer; at 0:00 screen turns red, alarm plays, POST /api/safety/missed-checkin triggers Twilio SMS to supervisor; show "Alert sent to [name]"
3. Dark theme, large touch targets for gloves
4. Store zone + check-in status in MMKV
5. Expo Notifications for incoming alerts

CHAPTER 09

Build Prompts — Week 2: Safety

Safety and maintenance modules

PROMPT 6 — SAFEPERMIT BYPASS MANAGEMENT

Build SafePermit (IEC 61511 compliant):

Backend: POST /api/bypasses/request, /api/bypasses/{id}/approve, GET /api/bypasses/active, POST /api/bypasses/{id}/close, GET /api/bypasses/risk-check

Risk matrix: input active bypass tags, detect dangerous combinations (e.g. 3 related ESD bypasses = CRITICAL), return risk_level + reason

Auto-expiry: Celery task every minute, SMS alert if bypass expired unclosed

Frontend: live dashboard of active bypasses as countdown cards with risk colour; Request New Bypass modal; risk alert banner; 30-day audit trail table

PROMPT 7 — EQUIPMENTIQ PREDICTIVE MAINTENANCE

Build EquipmentIQ with ML:

1. CUSUM drift detection on vibration/temperature/pressure
2. Isolation Forest (scikit-learn) trained on 30 days normal data, anomaly score 0-1
3. Linear regression trend predicting days to alarm limit
4. RUL bearing wear model

API: POST /api/equipment/reading, GET /api/equipment/{tag}/health, GET /api/equipment/alerts, POST /api/equipment/sound-analysis

Sound analysis: librosa FFT, extract fundamental + harmonics, compare to failure signatures (cavitation, bearing), return diagnosis + confidence + days_to_failure

Frontend: colour-coded asset health dashboard + trend charts

PROMPT 8 — ALARMIQ ALARM ANALYTICS

Build AlarmIQ (ISA-18.2 compliant):

Engine (pandas): input Exaquantum CSV (timestamp, tag, alarm_type, priority, acknowledged, duration); calculate alarm rate per hour, flood events (>10 in 10 min), stale alarms (>24h), chattering (>3/hr same tag), top 10 bad actors, shift breakdown; heatmap matrix hour vs day

API: POST /api/alarms/upload, GET /api/alarms/analysis/{id}, GET /api/alarms/report/{id}/pdf (ReportLab)

Frontend: drag-drop CSV upload; KPI cards with RAG status vs ISA targets; top 10 bad-actor bar chart; D3 heatmap; stale alarms table; ISA-18.2 PDF report button; recommendations

PROMPT 9 — CALTRACK CALIBRATION MANAGEMENT

Build CalTrack for I&C engineers:

Backend: instrument CRUD; POST /api/calibration/record auto-calculates drift_percent and next_due_date, generates ReportLab PDF certificate; GET /api/calibration/overdue, /api/calibration/due-soon, /api/calibration/{id}/trend; SMS alerts for overdue

Drift prediction: linear regression predicting when instrument exceeds plus/minus 2% limit

Frontend web: instrument list with filter, due-date calendar, drift trend charts

Frontend mobile: field form, scan QR, enter as-found/as-left, download PDF certificate

Seed 30 realistic IFFCO instruments (EJX pressure transmitters, temperature elements, flow/level transmitters)

CHAPTER 10

Build Prompts — Week 3: Intelligence

AI intelligence, prediction, and advanced features

PROMPT 10 — INCIDENTAI PATTERN PREDICTOR

Build IncidentAI:

- Incident DB seeded with IFFCO incidents and pre-incident signatures:
 - Dec 2020 pump P-201: bearing_temp +1.5C/hr, vibration_rms >4.2 mm/s, lube_oil_pressure dropping
 - Mar 2021 boiler: steam_pressure fluctuating, feed_water erratic, O2 dropping
 - Ammonia leak: NH3 suction pressure rising, seal_gas dropping, bearing_temp elevated
 - Pattern matching: extract current 72-hr trends, cosine similarity vs each signature, return top 3 with score, alert if >60%
 - Claude explanation in Hindi when >60%, suggest preventive actions
- API: GET /api/incidents/risk-assessment
- Frontend: risk panel showing similarity + Claude explanation

PROMPT 11 — 3D DIGITAL TWIN

Build 3D Plant Digital Twin in apps/web (Three.js + React Three Fiber):

- Dark scene, ambient + directional + green point lights, OrbitControls, green grid floor
 - Equipment as geometric shapes: ammonia reactor (tall cylinder), urea reactor, compressors (boxes with rotation), distillation towers, heat exchangers, pumps, Horton spheres (NH3 storage), connecting pipes
 - Live overlays via WebSocket: floating text temp/pressure/status; green/amber/red colour; pulsing glow on anomaly
 - Click equipment: side panel with name + tag, parameters, health score, 24-hr trend chart, Claude AI comment
 - Zone overlay toggle for hazard boundaries
- Make it the visual showstopper.

PROMPT 12 — ENERGYIQ OPTIMIZER

Build EnergyIQ:

- Aggregate from TimescaleDB: gas_flow, steam HP/MP/LP, power, urea_rate; calculate total Gcal/MT (target 5.200); hourly aggregates
 - Optimization via Claude every 4 hours: send 24-hr data, ask for 3 specific setpoint changes as JSON with parameter, current, suggested, saving_gcal, monthly_saving_rupees
- API: GET /api/energy/current, /api/energy/trend, /api/energy/suggestions, /api/energy/report
- Frontend: hero metric Gcal/MT vs target; breakdown donut (gas/steam/power); 30-day trend with target line; AI suggestion cards with Rs/month savings

PROMPT 13 — SHIFTLOG + AUTO REPORT

Build Digital ShiftLog (replaces paper notebook):

- Shifts A/B/C with auto-create at shift start; POST /api/shifts/entry (types: ALARM/MAINTENANCE/INCIDENT/OBSERVATION/PRODUCTION/EQUIPMENT); GET /api/shifts/current, /api/shifts/handover
 - POST /api/shifts/{id}/generate-report: ReportLab PDF with shift details, production summary, entries by type, equipment under maintenance, pending actions
 - WhatsApp delivery of PDF to supervisor
 - AI summary via Claude at shift end: 3 bullets Hindi + 3 English
- Frontend web: chronological feed + entry form + handover; mobile: quick entry with large type buttons
- Pre-populate realistic IFFCO shift data

CHAPTER 11

Build Prompts — Week 4: Deploy

Polish, deployment, and demo preparation

PROMPT 14 — DOCKER FULL STACK DEPLOYMENT

Create Docker Compose for DRISHTI full stack:
Services: postgres (TimescaleDB pg16), redis 7, api (FastAPI port 8000), web (Next.js port 3000), celery-worker, celery-beat, nginx (routes / to web, /api to api, /ws to api)
Create: apps/api/Dockerfile, apps/web/Dockerfile, nginx/nginx.conf, .env.example (ANTHROPIC_API_KEY, TWILIO_*, ELEVENLABS_API_KEY, DATABASE_URL, REDIS_URL), docker-compose.override.yml for hot reload, README with 3-command setup
GitHub Actions .github/workflows/deploy.yml: build images, run tests, deploy web to Vercel and API to Railway

PROMPT 15 — WEBSOCKET REAL-TIME SYSTEM

Build the real-time WebSocket system:
1. WebSocket manager: multiple connections, room-based per plant/zone, broadcast events (process_update, alarm_triggered, bypass_created, sos_activated, check_in_missed)
2. DCS simulator: APScheduler every 5s generating realistic IFFCO tags (UREA-TIC-201 150-165C, NH3-PIC-101 145-155 bar, CO2-FIC-102, PRILL-TIC-601, STM-PIC-701); 1% chance anomaly injection; save TimescaleDB + broadcast
3. Frontend hook useRealtimeData: connect ws://localhost:8000/ws/plant/phulpur, auto-reconnect, update Zustand store, LIVE indicator, real-time KPI updates without refresh

PROMPT 16 — DEMO MODE + DEMO DATA

Create demo mode for IFFCO engineer presentation:
1. Scenario controller: normal_operations, pump_anomaly_developing (P-201 bearing rising, IncidentAI 68% match), alarm_flood (15 alarms in 2 min), sos_activation; POST /api/demo/scenario/{name}
2. Demo seed: users R.K. Sharma (engineer) + D.K. Mishra (supervisor); 30 instruments with 90-day calibration history; 14 days shift logs; 48-hr process data with anomaly; 5 historical incidents; 3 resolved + 1 active bypass
3. DEMO.md flow guide: overview dashboard -> show printer photo -> ShiftLog -> PlantGPT Hindi query -> pump anomaly IncidentAI -> mobile Dead Man's Switch -> AlarmIQ CSV -> 3D twin -> closing line

CHAPTER 12

Engineer Demo Script

The exact 15-minute presentation for the head engineer at IFFCO Phulpur

Before the meeting: Confirm the paper shift register, printer usage, and alarm count on the HIS screen during your internship. Use your photos as evidence. Run the demo on your laptop. Load real IFFCO data (even a single week of alarm CSV from Exaquantum).

Opening (2 minutes)

"Sir, mujhe ek cheez dikhani hai. December 2020 mein is plant ke do engineers ki jaan gayi. March 2021 mein do aur. Maine ek system banaya hai jo shayad un charo engineers ko bacha sakta tha. Kya aap 15 minutes de sakte hain?"

[Show Overview Dashboard] "Ye DRISHTI hai — aapke Yokogawa DCS ke upar ek AI layer. Koi hardware nahi chahiye, koi OPC connection abhi nahi chahiye. Ye aapki existing data se kaam karta hai."

Problem 1 — Paper (2 minutes)

[Show your photo of the "For Daily Reports" printer and the green notebook] "Sir, ye aapke control room ka hi printer aur register hai — June mein maine photo li. Daily data paper pe jaata hai jabki DCS ke paas poori information electronically hai."

[Open ShiftLog] "DRISHTI mein operator yahi data yahan fill karta hai. Ek click mein PDF report ban jaati hai aur supervisor ke WhatsApp pe chali jaati hai. Koi printer nahi, koi paper nahi."

Problem 2 — Hindi AI (3 minutes)

[Open PlantGPT] "Yokogawa English mein hai. Hamara field technician Hindi mein bol sakta hai." [Speak] "Aaj raat kya problem aa sakti hai?" [Show Hindi response] "Ye aapke plant ka real data dekh kar predict kar raha hai."

Problem 3 — The Dec 2020 incident (4 minutes)

[Show IncidentAI] "December 2020 mein P-201 pump ke tie rods toot gaye. Vibration data exist karta tha — sirf koi analyze nahi kar raha tha." [Show anomaly: "68% match to Dec 2020 pattern"] "Ye alert 2 weeks pehle aata. Dec 2020 aur March 2021 — dono times. Woh charo log shayad aaj bhi yahan hote."

Problem 4 — Field safety (2 minutes)

[Show mobile Dead Man's Switch] "Agar koi engineer night shift mein alone hazardous zone mein jaata hai, ye timer start hota hai. 15 minute mein check-in nahi — aapke phone pe call aati hai."

Closing ask (2 minutes)

"Sir, main ek intern hoon. Maine ye 4 hafte mein banaya. Mujhe chahiye: (1) aapka guidance — kya ye problems IFFCO ke liye valuable hain? (2) ek Exaquantum alarm CSV real data pe test karne ke liye. (3) agar aap approve karein — internship ke baad IFFCO ke saath pilot. Ye eventually sare 5 plants pe deploy ho sakta hai."

CHAPTER 13

Business Model & Licensing

Turning DRISHTI into a real company

PHASE	WHO	PRICING	ANNUAL REVENUE
Pilot (now)	IFFCO Phulpur	Free — proof of concept	Case study + credibility
Phase 2 (3 months)	5 IFFCO plants	Rs 5 lakh / plant / year	Rs 25 lakh ARR
Phase 3 (12 months)	20 Indian urea plants	Rs 8 lakh / plant / year	Rs 1.6 crore ARR
Phase 4 (24 months)	Gulf + global	Rs 50 lakh / plant / year	Rs 5 crore+ ARR

Register before demo: One Person Company (OPC) registration costs Rs 5,000-10,000. Do this BEFORE presenting to IFFCO management. All IP must be in your company name. Suggested name: "Drishti Industrial Intelligence Pvt Ltd."

Protect your IP: Do NOT give IFFCO your source code. Provide a running SaaS license only. Your code stays yours; they pay annually for access. This is how you build value that can be sold or funded.

CHAPTER 14

CLAUDE.md — Project Context

Paste this as CLAUDE.md in your project root — Claude Code reads it automatically

CLAUDE.MD — COMPLETE PROJECT CONTEXT

```
# DRISHTI — Industrial AI Safety Platform

## What this project is
DRISHTI is an AI-powered industrial safety and intelligence platform for IFFCO Phulpur — an Ammonia and Urea facility in Uttar Pradesh, India, running Yokogawa CENTUM VP DCS. This platform runs ABOVE the DCS; it does not replace it.

## Core mission
Four engineers died at IFFCO Phulpur in 2020-2021. DRISHTI is the system that could have prevented those deaths. Every feature traces to a real, documented incident or confirmed problem.

## The stack
Monorepo: Turborepo + pnpm
apps/web: Next.js 15 + React 19 + TypeScript + Tailwind 4 + shadcn/ui + Three.js
apps/mobile: React Native 0.75 + Expo SDK 53 + Expo Router
apps/api: FastAPI + Python 3.12 + SQLAlchemy async + Celery + Redis
Database: TimescaleDB + PostgreSQL + Redis + Pinecone
AI: Claude (claude-sonnet-4) + Whisper v3 + ElevenLabs + scikit-learn + PyTorch + YOLOv8 + librosa
Infra: Docker Compose + GitHub Actions + Vercel + Railway

## Design principles
- Dark industrial theme: background #030806, primary green #00C864
- LARGE touch targets — engineers wear gloves
- HINDI first, then English — every user-facing string supports both
- OFFLINE capable — mobile must work in poor connectivity
- REAL-TIME — WebSocket push for all live data

## Key industrial terms
DCS = Distributed Control System; HIS = Human Interface Station; FCS = Field Control Station; SIS/ESD = Safety Instrumented System / Emergency Shutdown; Bypass = temporarily disabling a safety interlock; HART = smart instrument protocol; OPC-UA = modern data standard; Exaquantum = Yokogawa historian; Gcal/MT = energy KPI; TAG = instrument ID (NH3-FIC-101); As-found/As-left = calibration values; ISA-18.2 = alarm management standard

## User roles
ENGINEER (mobile), OPERATOR (web dashboard), SUPERVISOR (approves permits, gets alerts), MANAGER (KPI dashboard)

## Never do
- Never connect to real DCS without explicit authorization
- Never expose raw OPC/DCS credentials
- Use mock/simulated data for development, labeled clearly as DEMO DATA
- Never store sensitive plant data outside an encrypted database

## Demo scenario
Pump anomaly where P-201 bearing temp rises slowly and IncidentAI flags 68% similarity to the Dec 2020 incident. This is the key wow moment.
```

You now have everything. 40 sourced problems, a complete solution map, the full 2026 tech stack, system architecture, every feature spec, a 4-week build plan, 16 copy-paste build prompts, the engineer demo script, the business model, and the project context file. Start with Chapter 8, Prompt 1. Build one prompt per day. In 30 days, DRISHTI is real.